Sri Lanka Institute of Information Technology

Programming Applications and Frameworks (IT3030) - Initial Document



Our Team

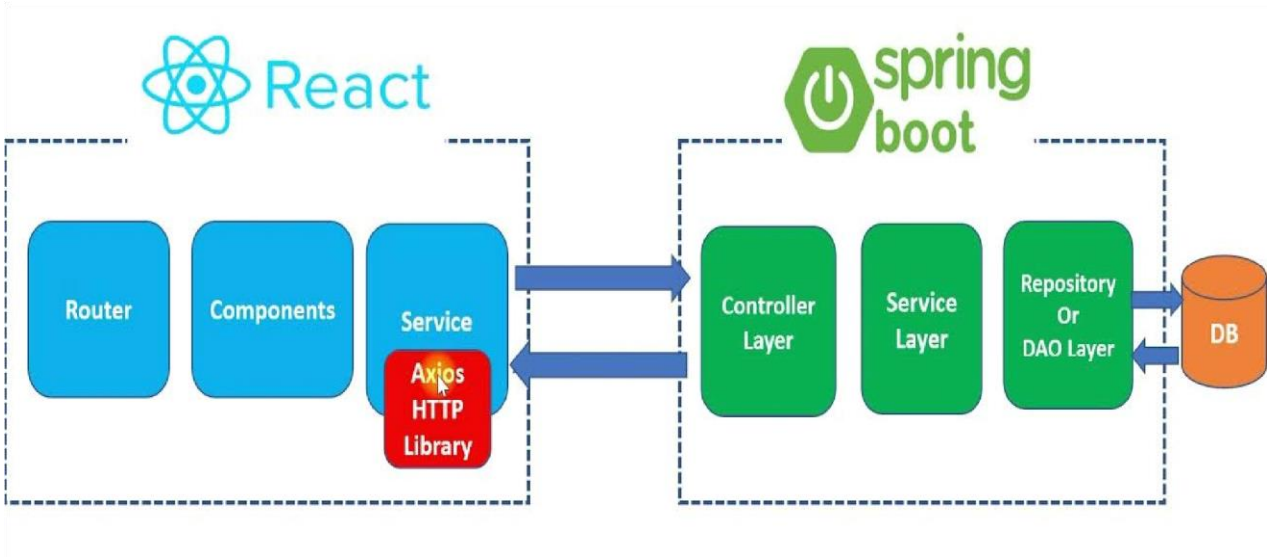
- N T B P Dias - IT22134080
- M T K Chathuri - IT22548078
- D R Dissanayaka - IT22207654
- S D Nirmitha - IT22213976

Coding Skill-Sharing & Learning Platform

System Overview

The Skill-Sharing & Learning Platform for Coding is a web-based application designed to enable users to share and learn various coding skills. The platform allows users to upload coding tutorials, share progress in their coding journeys, create structured learning plans. This initiative fosters a collaborative and engaging environment for individuals passionate about coding. The project will be developed using a Java (Spring Boot) REST API for the backend and a React based web application for the frontend. The system will be version-controlled using Git and hosted on GitHub, with GitHub Workflow managing CI/CD pipelines.

Diagram



Functional Requirements

1. **User Profiles & Social Features**
   - Public profile pages displaying shared posts, activities, and coding achievements.
   - Follow other users to view their coding-related updates and posts.
2. **Coding Skill Sharing Posts**
   - Users can upload coding tutorials or examples.(text, images, or videos up to 30 seconds)
   - Add detailed descriptions to shared content.
3. **Learning Progress Updates**
   - Users can post updates on their coding progress using predefined templates.
   - Share milestones such as completed projects, solved challenges, or new concepts learned.
4. **Learning Plan Sharing**
   - Create, view, and update structured coding learning plans.
   - Include topics, resources (e.g., links to GitHub repositories or coding platforms), and timelines.
5. **Interactivity & Engagement**
   - Like and comment on coding posts.
   - Edit or delete comments.
6. **Notifications**
   - Notify users of likes and comments on their posts.
7. **Authentication**
   - Secure login via OAuth 2.0 using social media accounts.

Non-Functional Requirements

1. **Performance**
   - Ensure quick response times for API requests
   - Optimize media uploads and retrieval for coding snippets and tutorials.
2. **Scalability**
   - Design a scalable system to accommodate a growing user base and increased content uploads.
3. **Security**
   - Use HTTPS for secure data transmission and Implement authentication with OAuth 2.0.
4. **Usability**
   - Provide an intuitive interface for coding enthusiasts of all skill levels.
   - Ensure accessibility with responsive design.
5. **Availability**
   - Ensure 99.9% uptime through reliable hosting and monitoring.

Main Functions

1. **Authentication & User Management (N T B P Dias - IT22134080)**

Description:

- This function focuses on user authentication and profile management.
- It ensures secure access to the platform using OAuth 2.0 for login through social media accounts like Google or Facebook.
- Users can set their privacy preferences (e.g., public or private profile)

CRUD Operations:

Create: Register a new user and create their profile with basic details (name, email, bio, skills)

Read: View user profile details, including bio, skills, interests, and uploaded media (profile picture and cover photo).

Update: Modify profile information, such as updating the bio, adding new skills, or changing privacy settings.

Delete: Remove the user profile permanently from the platform.

2. **Coding Posts Management (M T K Chathuri - IT22548078)**

Description:

- This function allows users to share coding tutorials with the community.
- Users can upload media files (3 images or 30-second videos) and provide descriptions.
- Posts can be liked and commented on by others

CRUD Operations:

Create: Users can create new posts by sharing coding tutorials or achievements including the description.
Read: View skill-sharing posts from other users, including photos/videos and descriptions.
Update: Users can edit their posts to improve descriptions, replace media files.
Delete: Users can remove outdated or irrelevant posts to keep their shared content current.

3. **Learning Plans Management (D R Dissanayaka - IT22207654)**

Description:

• Users can create structured learning plans to track progress in learning coding skills.
• Learning plans can include topics, resources Add topics, resources, and completion timelines.

CRUD Operations

Create: Users can create new learning plans, defining topics, resources, and timelines to structure their coding journey.
Read: Users can view their existing learning plans, providing a clear overview of their progress and goals.
Update: Users can edit or modify learning plans to reflect changes in topics, resources, or timelines.
Delete: Users can remove outdated or irrelevant learning plans to maintain an organized list.

4. **Learning Progress Updates Features (S D Nirmitha - IT22213976)**

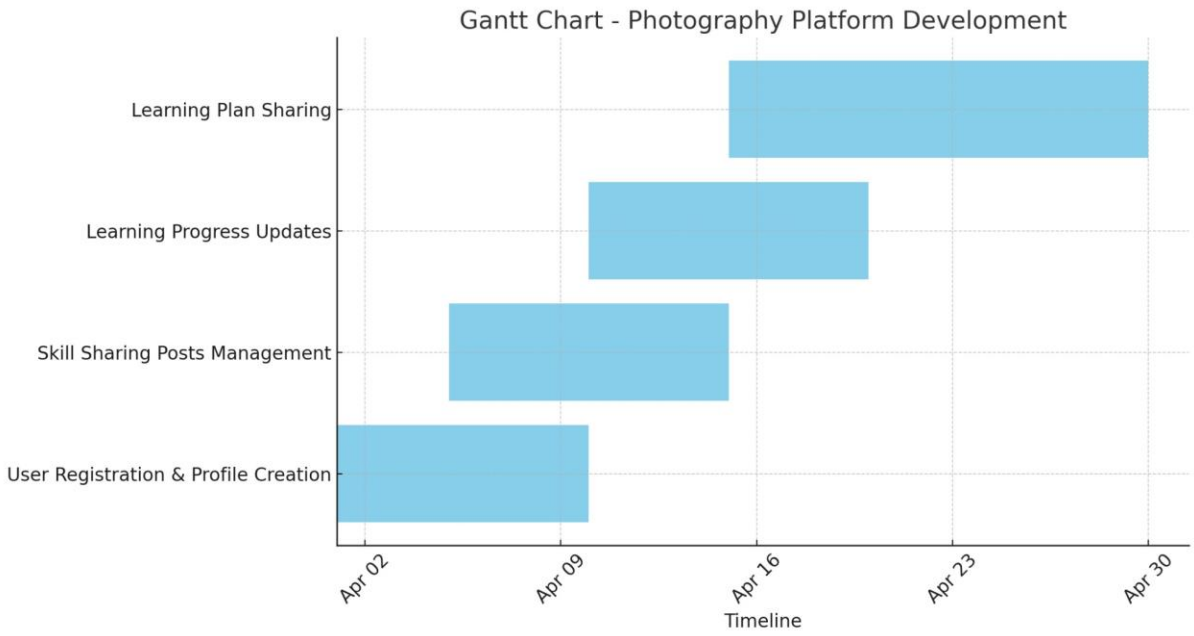
Description

● Users can post learning journey updates
● Predefined templates for structured updates (e.g., completed tutorials, new skills learned)
● Users can like and comment on updates

CRUD Operations

Create: Users can create new learning progress posts using the structured templates.
Read: Users can view learning updates posted by others, allowing them to stay informed about their peers' progress and gain inspiration or insights.
Update: Users can edit or modify their progress posts if they need to add more details, correct mistakes, or update their achievements.
Delete: Users can remove old or irrelevant updates to keep their learning journey feed clean and up-to-date.

Gantt Chart


Gantt Chart - Photography Platform Development
Learning Plan Sharing
Learning Progress Updates
Skill Sharing Posts Management
User Registration & Profile Creation
Apr 02
Apr 09
Apr 16
Apr 23
Apr 30
Timeline